

Given:

- $\Delta t = 1 \text{ sec}$
- $\alpha_0 = 30$
- Initial:
 - Car-1: $v_0 = 0, x_0 = 0$
 - Car-2: $v_0 = 0, x_0 = -25$

Leading vehicle (Car-1)

Acceleration:

- $t = 1 \text{ to } 7 \rightarrow a = 4$
- $t = 8 \text{ to } 10 \rightarrow a = -2$

Speed (v)

$$v(t+1) = v(t) + a$$

Position (x)

$$x(t+1) = x(t) + v(t) + 0.5a$$

Following vehicle (Car-2)

Acceleration:

$$a(t+1) = \alpha_0 / (x_1 - x_2) \times (v_1 - v_2)$$

t = 0

- Car-1: $v=0, x=0$
- Car-2: $v=0, x=-25$

t = 1

Car-1:

- $v_1 = 0 + 4 = 4$
- $x_1 = 0 + 0 + 2 = 2$

Car-2:

- $a_1 = 30 / (0 - (-25)) \times (0 - 0) = 0$

- $v_1 = 0$
- $x_1 = -25$

t = 2

Car-1:

- $v_2 = 4 + 4 = 8$
- $x_2 = 2 + 4 + 2 = 8$

Car-2:

- $a_2 = 30 / (2 - (-25)) \times (4 - 0)$
- $a_2 = 30 / 27 \times 4 = \mathbf{4.44}$
- $v_2 = 0 + 4.44 = 4.44$
- $x_2 = -25 + 0 + 2.22 = -22.78$

t = 3

Car-1:

- $v_3 = 8 + 4 = 12$
- $x_3 = 8 + 8 + 2 = 18$

Car-2:

- $a_3 = 30 / (8 - (-22.78)) \times (8 - 4.44)$
- $a_3 = 30 / 30.78 \times 3.56 \approx \mathbf{3.47}$
- $v_3 = 4.44 + 3.47 = 7.91$
- $x_3 = -22.78 + 4.44 + 1.74 = -16.60$

t = 4

Car-1:

- $v_4 = 12 + 4 = 16$
- $x_4 = 18 + 12 + 2 = 32$

Car-2:

- $a_4 = 30 / (18 - (-16.60)) \times (12 - 7.91)$
- $a_4 = 30 / 34.60 \times 4.09 \approx \mathbf{3.55}$
- $v_4 = 7.91 + 3.55 = 11.46$
- $x_4 = -16.60 + 7.91 + 1.78 = -6.91$

t = 5

Car-1:

- $v_5 = 16 + 4 = 20$
- $x_5 = 32 + 16 + 2 = 50$

Car-2:

- $a_5 \approx 30 / (32 - (-6.91)) \times (16 - 11.46)$
- $a_5 \approx 30 / 38.91 \times 4.54 \approx \mathbf{3.50}$
- $v_5 = 11.46 + 3.50 = 14.96$
- $x_5 = -6.91 + 11.46 + 1.75 = 6.30$

Time (sec)	Leading Vehicle			Following Vehicle		
	Acceleration (ft/s ²) - a	Speed (ft/s) - v	Distance (ft) - x	Acceleration (ft/s ²) - a	Speed (ft/s) - v	Distance (ft) - x
Time	a ₁ (ft/s ²)	v ₁ (ft/s)	x ₁ (ft)	a ₂ (ft/s ²)	v ₂ (ft/s)	x ₂ (ft)
0	0.00	0.00	0.00	0.00	0.00	-25.00
1	4.00	4.00	2.00	0.00	0.00	-25.00
2	4.00	8.00	8.00	4.44	4.44	-22.78
3	4.00	12.00	18.00	3.47	7.91	-16.60
4	4.00	16.00	32.00	3.55	11.46	-6.91
5	4.00	20.00	50.00	3.50	14.96	6.30
6	4.00	24.00	72.00	3.40	18.36	22.08
7	4.00	28.00	98.00	3.20	21.56	41.40
8	-2.00	26.00	125.00	2.80	24.36	64.08
9	-2.00	24.00	150.00	2.30	26.66	89.05
10	-2.00	22.00	173.00	1.80	28.46	115.34